DE–5867 **13**

DISTANCE EDUCATION

B.Sc. (CS) DEGREE EXAMINATION, MAY 2013.

DIGITAL COMPUTER FUNDAMENTALS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Convert the following decimal numbers to Hexadecimal form

   (i) 8234

   (ii) 525.225. (10)

   (b) Explain binary addition and subtraction with example. (10)

2. (a) Simplify the following boolean function in product-of-sume form by means of a four-variable map

   $F(w,x,y,z) = \sum(2,3,4,5,6,7,11,14,15)$. (10)

   (b) Explain in detail about Decoders. (10)

3. (a) Explain the Full-Adder with circuit diagram and truth table. (10)

   (b) Explain the addition and subtraction in a parallel arithmetic element. (10)

4. (a) Explain in detial about Gated Flip-flops. (10)

(b) Explain the Binary counters. (10)

5. (a) Discuss about the Input-output units. (10)

(b) Write about the different types of memory. (10)

6. (a) With example, explain subtraction with complements. (10)

(b) Convert the following :

(i) $(349)_8 = (\ )_{10}$

(ii) $(243.265)_8 = (\ )_{10}$. (10)

7. (a) Explain the multiplexors with the logic diagram and the function table. (10)

(b) Write about Encoders with a neat diagram. (10)

8. (a) Explain the types of shift registers. (10)

(b) Write briefly about the generations and classification of computers. (10)

DE–5868 **14**

DISTANCE EDUCATION

B.Sc. (CS) DEGREE EXAMINATION, MAY 2013.

DATA STRUCTURES USING 'C'

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the various data types available in 'C' with example. (10)

   (b) Explain all formatted and unformatted input and output statements. (10)

2. (a) Explain the following with example

   (i) Switch

   (ii) Break

   (iii) Continue. (8)

   (b) Discuss on the various loop statements in 'C'. (12)

3. (a) Define Array. Write a 'C' program for arranging n numbers in ascending order. (10)

   (b) What is recursion? Write a 'C' program for calculating factorial of given number using recursion. (10)

4. (a) What is structure in 'C'? Explain with example. (10)

   (b) Explain the various file operations in 'C' with example. (10)

5. (a) What is stack? Explain the operations related with stack. (10)

   (b) Define queue. How to insert a data item in a queue? (10)

6. (a) What is binary tree? Explain binary tree representation with example. (10)

   (b) Explain the differences between singly and doubly linked list. (10)

7. (a) Explain the applications of tree. (10)

   (b) Explain the evaluation of expression using stack. (10)

8. (a) Explain the differences between union and structure. (10)

   (b) Write a 'C' program for creating a student file. (10)

DE–5869 **15**

DISTANCE EDUCATION

B.Sc. (CS) DEGREE EXAMINATION, MAY 2013.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions. (5 × 20 = 100)

All questions carry equal marks.

1. (a) Construct the truth table of $(Q \wedge (P \to Q) \to P)$. (7)

   (b) Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a tautology. (7)

   (c) Write an equivalent expression for $(P \Leftrightarrow Q \wedge R) \vee (R \Leftrightarrow S)$ which contains neither the biconditional nor the conditional. (6)

2. (a) Show that $A \subseteq B \Leftrightarrow NB \subseteq NA$ (7)

   (b) Let $X = \{1,2,....7\}$ and $R = \{(x,y) / x - y \text{ is divisible by } 3\}$. Prove that R is an equivalence relation. (6)

   (c) Let R and S be two relations on a set of positive integer $I : R = \{<x, 2x> / x \in I\}$ $S = \{<x, 7x> / x \in I\}$ Find $R \circ S, R \circ R$ and $R \circ R \circ R$. (7)

3. (a) Define bijective function with an example. (7)

   (b) Let * be a binary operation on X which is associative and which has the identity $e \in X$. If an element $a \in X$ is invertible, then prove that both its left and right inverses are equal. (7)

(c) Show that the function $f(x,y)=(x+y)$ is primitive recursive function. Hence compute the value of $f(2,4)$. (6)

4. (a) Define monoid and semigroup. (7)

(b) Prove that for any commutative monoid $<M,*>$, the set of idempotent elements of M forms a submonoid. (7)

(c) Prove that a subset $S \neq \phi$ of G is a subgroup of $<G, *>$ if and only if for any pair of elements $a,b \in s, a*b^{-1} \in s$. (6)

5. (a) State and prove that the fundamental theorem of group homomorphisms. (10)

(b) Define (10)

(i) Normal subgroup

(ii) Direct product

(iii) Kernel

(iv) Permutation.

6. (a) Prove that in any graph, the number of odd degree vertices is even. (7)

(b) Draw diagraph G corresponding to adjacent matrix (7)

$$A = \begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \end{array} \begin{array}{c} \begin{array}{cccc} V_1 & V_2 & V_3 & V_4 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

(c) Prove that a tree with n vertices has (n-1) edges. (6)

7. (a) Define $(5 \times 2 = 10)$

(i) Pendent vertex

(ii) Null graph

(iii) Nodes of the graph

(iv) Bipartite graph

(v) Regular graph.

(b) Prove that a graph G without loops is a tree if and only if any two distinct vertices are connected by a unique path. (10)

8. (a) Let $f(x)=x+2$, $g(x)=x-2$ and $h(x)=3x$ for $x \in R$, where R is the set of real numbers. Find $g \circ f, f \circ g, f \circ f, g \circ f \circ h, f \circ h$. (7)

(b) Prove that for any two sets A and B $A-(A \cap B)=A-B$. (7)

(c) Define quantifier and negation of quantifier statement. (6)

DE–5870 **23**

DISTANCE EDUCATION

B.Sc. (Computer Science/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the basic concepts of object oriented programming. (10)

   (b) Explain the different types of operators in C++. (10)

2. (a) Explain access specifier with example. (10)

   (b) Explain friend function with example. (10)

3. (a) Explain : (10)

   (i) this pointer

   (ii) pointer to members.

   (b) Explain dynamic constructor with example. (10)

4. (a) Explain function overloading with example. (10)

   (b) With a program explain overloading of unary operators. (10)

5. (a) Explain the different types of inheritance. (10)

   (b) How runtime polymorphism is achieved in C++? (10)

6. (a) Write a program in C++ to implement the concept of virtual function. (10)

   (b) Discuss about constructors and destructors with example. (10)

7. (a) Write a C++ program that overleads the + and = = operators to concatenate and compare two strings respectively. (10)

   (b) Explain : (10)

      (i) Cin and Cout streams

      (ii) Manipulators.

8. (a) Explain any five string manipulation functions. (10)

   (b) Explain inline function with suitable example. (10)

DE–5871 **24**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve

$$x + 3y + 3z = 16$$
$$x + 4y + 3z = 18$$
$$x + 3y + 4z = 19$$

by Gauss - Jordan method. (10)

(b) Explain the Gauss elimination method for solving linear systems of the form $AX = B$. (10)

2. (a) Apply Gauss-seidal method to solve the system of equations. (10)

$$20x + y - 2z = 17$$
$$3x + 20y - z = -18$$
$$2x - 3y + 20z = 25$$

(b) Find the value of y at $x = 21$ from the following data. (10)

x :	20	23	26	29
y :	0.342	0.3907	0.4384	0.4848

3. (a) Use Lagrange's formula to fit a polynomial to the data (10)

x :	–1	0	2	3
y :	–8	3	1	12

(b) Explain about Newton's forward and backward interpolation. (10)

4. Dividing the range into ten equal parts, find the approximate value of

$\int_0^{\pi} \sin x \, dx$ by

(a) Trapezoidal rule

(b) Simpson's 1/3 rule. (10 + 10)

5. (a) Mean of 100 items is found to be 30. If at the time of calculation two items are wrongly taken as 32 and 12 instead of 23 and 11. Find the correct mean. (10)

(b) Calculate mean deviation of the following : (10)

X :	10	11	12	13	14
Y :	3	12	18	12	3

6. (a) A bag contains 7 red balls and 5 white balls. 4 balls are drawn at random. What is the probability that

(i) all of them are red

(ii) two of them are red and two white. (10)

(b) An article manufactured by a company consists of two parts A and B. In the process of manufacture of part A, 9 out of 100 are likely to be defective. Similarly 5 out of 100 are likely to be defective in the manufacture of part B. Calculate the probability that the assembled part will not be defective. (10)

7. (a) The increase in weights due to two kinds of food are given below. Can it be said that food B is better than A? (10)

Food A :	49	53	51	52	47	50	52	53
Food B :	52	55	52	53	50	54	54	53

(b) Applying χ^2 test, with the help of the following data find out whether there is any relationship between smoking and drinking. (10)

	Drinking	Not-drinking
Smoking	74	26
Not-smoking	10	30

8. The three samples below have been obtained from normal population with equal variances. Test the hypothesis at 5% level that the population means are equal. (20)

8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

DE–5872 **25**

DISTANCE EDUCATION

B.Sc. (Computer Science) / B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum :100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Briefly explain the following
   (a) Windows Desktop (5)
   (b) Office shortcut bar (5)
   (c) Menu commands (5)
   (d) Dialog boxes (5)

2. (a) How can you open a document in MS-office? (5)
   (b) Explain various steps of saving the document in first time and Re-saving a document. (5)
   (c) Write short notes about Toolbars. (10)

3. (a) What is called a template? Explain different types of templates in word. (5)
   (b) How can you Merging a letter with data? (10)
   (c) What is the use of Header and Footer? (5)

4. (a) How can you print a document? (5)

   (b) What is the use of Mailmerge? (5)

   (c) How can you Align and Indent the given text? (5)

   (d) Write about the Formatting text. (5)

5. (a) Explain the steps involved in chart insertion in MS Excel. (5)

   (b) How can you find and replace data in MS Excel? (10)

   (c) Explain fill and clear option in MS Excel. (5)

6. (a) Describe the importance of Power point and list out its various advantages. (5)

   (b) How can you Add and Delete a slide in MS-POWER POINT? (5)

   (c) Write short notes about the graph chart and organization chart. (10)

7. Write short notes on

   (a) Moving from Slide to Slide in MS-POWER POINT.(5)

   (b) Drawing on Slides in MS-POWER POINT. (5)

   (c) Adding speaker notes in MS-POWER POINT. (10)

8. (a) Explain the features of MS Access and what are its purposes? (5)

   (b) Explain database creation using wizard option in MS Access? (10)

   (c) How can you adjust the column width and Height in MS Access? (5)

DE–5873 **31**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) When graphics is said to be interactive and non interactive. (5)

   (b) How light pen acts as a good pointing device? (5)

   (c) Describe an algorithm for circle generation. (10)

2. (a) Mention the points to be kept in mind for designing a good graphics system software. (6)

   (b) Describe the Bresenham's algorithm for generating straight lines. (6)

   (c) What are the components in a cathode Ray tube? How it works? (8)

3. (a) List the principles involved in a two dimensional transformation. (6)

   (b) Define clipping. (4)

   (c) Explain the concept of rotating two dimensional object (i) at origin (ii) at any point. (10)

4. (a) How to converts window coordinates into world coordinates? (5)

   (b) Give the importance of matrix representation of transformations. (5)

   (c) How clipping can be done for line segment? (10)

5. (a) How to represent a three dimensional object? (5)

   (b) How translation is done for a three dimensional object? (5)

   (c) Explain the concept of composite transformations. (10)

6. (a) How to perform the job of scaling with respect to three dimensional object? (6)

   (b) What is meant by mirror reflection? (4)

   (c) Derive the matrix representation for three dimensional objects. (10)

7. (a) How to construct a three dimensional view? (5)

   (b) Explain the concept of depth comparison. (5)

   (c) Describe a hidden surface algorithm. (10)

8. (a) Describe the construction concept of the user's model. (10)

   (b) Briefly discuss on the merits and demerits of the command language. (10)

DE–5874 **32**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Give a note on graphical user interface. (10)

   (b) Discuss the features of windows 95 in detail. (10)

2. (a) Narrate the procedure of selecting and sizing windows. (10)

   (b) Write about PIF editor. (10)

3. (a) Describe the concept of multitasking. (10)

   (b) Compare and contrast OLEDB with ODBC. (10)

4. Give a detailed description on integrated development environment of Visual Basic. (20)

5. (a) Narrate the procedure of creating slider controls and handling slider events. (10)

   (b) With an illustration, explain the concept of logical testing. (10)

6. (a) How will you make selections with Switch( ) and Choose ( ) functions? Explain with syntax. (10)

   (b) Explain the importance of forms and controls in Visual Basic. (10)

7. Elucidate the procedure of creating and displaying pop – up menus with syntax and suitable example. (20)

8. Describe the following concepts :

   (a) Loading, showing and hiding forms. (10)

   (b) Parts of an MDI form and arranging MDI child windows. (10)

DE–5875 **33**

DISTANCE EDUCATION

B.Sc. (C.S.)/B.Sc. (C.S.) (Lateral) DEGREE EXAMINATION, MAY 2013.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the evolution of operating systems briefly. (10)

   (b) List out the functions of operating system. (10)

2. (a) What is process? What are its states? Explain with state transition diagram. (10)

   (b) Discuss the elements of process control block. (10)

3. (a) Explain round robin scheduling algorithm in detail. (10)

   (b) What is meant by priority scheduling? Explain briefly. (10)

4. (a) State the necessary conditions for a deadlock. (10)

   (b) Explain how does a semaphore becomes a solution to critical section problem. (10)

5. (a) Explain the organization of input-output functions briefly. (10)

   (b) Write short notes on interrupt handlers. (10)

6. (a) Discuss in detail the fixed variable multiple partition memory management systems. (10)

   (b) Explain segmentation in memory management and bring out its advantages. (10)

7. (a) Explain segmentation with paging briefly. (10)

   (b) What is meant by virtual memory? Discuss briefly with an example. (10)

8. (a) Explain the file accessing methods briefly. (10)

   (b) Mention the security mechanisms to protect the files. Explain with an example. (10)

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DE–5876 **34**

DISTANCE EDUCATION

B.Sc. (C.S)/ B.Sc.(C.S) (Lateral) DEGREE EXAMINATION, MAY 2013

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) List the advantages and disadvantages of using a computer on a network. (10)

   (b) List the functions of each layer in the OSI model. (10)

2. (a) What are the different types of headers in an HTTP message from the client? (10)

   (b) Explain the uses of web server. (10)

3. (a) What is URL? How does the URL differ from the domain name? (10)

   (b) List out the features and services from e-mail programs. (10)

4. (a) Explain how to find a job at school using internet facility. (10)

   (b) Illustrate briefly about the entertainment on the internet. (10)

5. (a) How does internet is useful for doing business? Explain. (10)

   (b) Bring out the features of Electronic commerce with examples. (10)

6. (a) List out the unique features of Java programming language. (10)

   (b) What are the primitive data types in Java? Give examples. (10)

7. (a) Explain the method of overloading with examples. (10)

   (b) What is Java byte code? How do we get the byte codes from the source codes? Explain. (10)

8. (a) What is meant by constructor? Give example. (10)

   (b) Write short notes on string handling with examples. (10)

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DE–5877 **35**

DISTANCE EDUCATION

B.Sc. (C.S)/B.Sc (C.S) (Lateral)DEGREE EXAMINATION, MAY 2013

RDBMS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) List out the disadvantages of file processing system. (10)

   (b) Explain schema and subschema with an example. (10)

2. (a) Explain the Entity-Relationship Model. (10)

   (b) List out the features of data in a database. (10)

3. (a) Explain the data definition facilities briefly. (10)

   (b) Bring out the characteristics of Database approach. (10)

4. (a) What is meant by data control? Explain briefly. (10)

   (b) Discuss any five components of DBMS. (10)

5. (a) Explain various parts of SQL Languages briefly. (10)

   (b) Comment on working with NULL value with examples. (10)

6. (a) Explain the two primary structures of Oracle database. (10)

   (b) Illustrate the process of joining multiple Tables in a Query. (10)

7. (a) Illustrate how a table is created with an example. (10)

   (b) Describe in detail how the structure of table is modified with an example. (10)

8. Write short note on

   (a) privileges on a database. (10)

   (b) embedded SQL. (10)

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DE–6611 **16**

DISTANCE EDUCATION

B.Sc. (Computer. Science) DEGREE EXAMINATION,
MAY 2013.

C AND DATA STRUCTURES – LAB

Time : Three hours Maximum : 100 marks

Examiner has to select one question and give it to the students by lot system.

Answer both subdivisions (a) and (b).

1. (a) Write a C program to calculate compound interest.

   (b) Write a C program to implement PUSH and POP operations on stack.

---------- Cut here ----------

2. (a) Write a C program to sort 10 numbers, in ascending order.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

3. (a) Write a C program to find the factorial of a given number using FUNCTION declaration.

   (b) Write a C program to sort 10 numbers in Ascending order with naming of variable and the value before and after sorting.

4. (a) Write a C program to input a basic pay and calculate gross pay and Net pay.

Income :

HRA = 15% of BASICPAY

DA = 90% OF BASICPAY

GROSS PAY = BASICPAY + DA + HRA

DEDUCTIONS :

PF = 10% OF BASICPAY

WCHARGE = 200

DEDUCTIONS = PF + WCHARGE

NETPAY = GROSS PAY – DEDUCTIONS.

(b) Write a C program to sort a set of elements using Insertion sort.

---------- Cut here ----------

5. (a) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS' otherwise 'FAIL' . Write a C program to do the above.

(b) Write a C program to sort a set of elements using selection sort.

DE–6612 **26**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

LAB II — C++ PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.

1. (a) Write a menu driven C++ program to add and subtract given two matrices of order M × n defined in class, using operator overloading.

   (b) Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to t return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipts and issues updating the stock level as appropriate. Write a menu driven C++ program to solve the problem.

---------- Cut here ----------

2. (a) Using pointers create a class and write a program to get the n names and display them in sorted order.

   (b) Write a program using polymorphism to calculate the square of any two numbers of type int, float double and long.

3. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

(b) Write a program using friend function which can act on the classes alphas and beta. Using constructors fix the values for alpha and beta.

---------- Cut here ----------

4. (a) Using operator overloading, write a C++ program to find the different and total length of given two various tubes specified in meters and centimeters.

(b) Create a class int to overload all five integer arithmetic operators (+, –, *, / and %) so that they operate on objects of type int. If the result of any such arithmetic operation exceeds the normal range of int's from – 32,768 to 32767 – have the operator print a warning and terminate the program. Write a program to test this class.

---------- Cut here ----------

5. (a) Write a C++ program to reverse the sentence and find the given sentence is palindrome or not.

(b) Create a class called employee that contains a name and an employee number. Include a member function called get data ( ) to get data from the user, another function called put data ( ) to display the data. Write a main ( ) program to exercise this class. It should create an array of type employee and then invite the user to input data for n employees.

6. (a) Create a class DONOR that contains donor number, donor name, age, address, sex, blood group.

Write a menu driven C++ program to display the number, name and address of the donors of the following categories.

(i) blood donors having the blood group A+

(ii) blood donors in the age group between 16 to 25

(iii) female donors having blood group O+

(b) Write a C++ program which will accept a string of 10 characters in length from the key board and count the occurrences of each of the five vowels in the string.

DE–6613 **27**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

Lab III — APPLICATION PROGRAMS

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question each from WORD, EXCEL and ACCESS

MS-Word : 25
EXCEL : 35
ACCESS : 40
Total : 100

SECTION A

MS-WORD

1. Prepare the first page of your record note book with picture insertion and alignment.

---------- Cut here ----------

2. Type lecture notes on provide audio explanation with the help of sound files.

---------- Cut here ----------

3. Create a table of student data that contains REGNO, NAME, ENGLISH, MATHS and SCIENCE marks. Add a new column named TOTAL and the find the row total for each student.

---------- Cut here ----------

4. Illustrate the mail merge concept to apply for the post of programmer for at least five companies.

SECTION B

MS-EXCEL

5. Create a worksheet in Excel with the following columns :

   CONSNO, TYPE, NAME, PR, CR, UNITS, AMT.

   (a) Add data for 10 customers with type D for domestic user and type I for industrial user.

   (b) Calculate the bill using the slab. (Assume yourself).

---------- Cut here ----------

6. Create a worksheet with the following columns :

   SNO, SNAME, CITY, PNAME, SAMT.

   (a) Add 5 records for five different salesmen who have carried out sales of different products in different cities.

   (b) Find citywise and salesmanwise sales.

---------- Cut here ----------

7. The following were the observations made in certain experiments for the values y and given the values of x.

x :	1	2	3	4	5	6	7	8	9	10
y :	10	30	45	25	15	28	40	32	15	35

   Compute the following :

   (a) Find mean, median and standard deviation

   (b) Correlation coefficient between x and y

   (c) Draw a chart in Excel and paste it word.

---------- Cut here ----------

8. Create a worksheet for mark list of your class for each semester. (One worksheet per semester). Draw a graph to illustrate class performance - subjectwise. Also draw a pie chart for overall performance.

SECTION C

MS-ACCESS

9. Create an employee table with the fields.
EMPNO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY.
   (a) Add data for 10 employees
   (b) Write a query to display all male employees whose salary is between 10000 and 50000 and living in city "CHENNAI".
   (c) Write a query to display all female employees whose ages are in the range 20–30.

---------- Cut here ----------

10. Create a student table with the following fields :
REGNO, NAME, M1, M2, M3
Create an address table with the fields :
REGNO, STREET, CITY and PIN.
   (a) Write a query to display REGNO, NAME and total of all three subject marks.
   (b) Write a query to display REGNO, NAME, STREET, CITY, PIN and total of all three marks.

---------- Cut here ----------

11. Create inventory table with the fields :
ITEMNO, NAME, QUANTITY, ON HAND and REORDER LEVEL.
   (a) Create a form in custom format
   (b) Create a query REORDER to show all items
   (c) Create a macro that executes REORDER query automatically.

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12. Create two tables SALES1 and SALES2. Join them to produce a third table SALES3.

DE–6614 **36**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)

DEGREE EXAMINATION, MAY 2013.

LAB IV — VISUAL BASIC AND ORACLE

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and it is given to the candidate by Lot system

1. (a) Write a VB program to calculate simple interest or compound interest using three command buttons namely input, calculate and display.

(b) Create a table BLOOD with the following fields :

DNO, DNAME, DOB, DAGE, DADD1, DADD2, DCITY, DPIN, DBLOOD, DSEX.

(i) Add 6 records and display all the records.

(ii) Display the information of donors in the age group 16 to 20 using BETWEEN option.

(iii) Display the information of donors having any of the blood group from the set of blood groups using IN option.

2. (a) Write and test a VB program to select candidates for four posts based on their qualifications.

Male and PG degree	:	Manager
Male, graduate and typing	:	Clerk
Female and PG Degree	:	Secretary
Female, graduate and typing	:	Steno

(b) Create a table MARKS with the following fields :

REGNO, NAME, MARK1, MARK2, MARK3, TOTAL, AVG

(i) Add 5 records

(ii) Replace all total with MARK1 + MARK2 + MARK3 and display the contents.

(iii) Replace all AVG with TOTAL/3 and display the contents.

(iv) Display the maximum and minimum marks in each subject.

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3. (a) Write a VB program to do temperature conversion C to F and F to C at user's choice using label, text and enter key.

(b) Create a table LABEL with the following fields :

NAME, STREET, CITY, PIN

(i) Add 10 records

(ii) Display the contents of the table

(iii) Display the ascending order sorted list with NAME as primary key

(iv) Display the contents with appropriate headings.

4. (a) Write and test a VB program to compute and print either the sum or the PRODUCT of the first N natural numbers. Use option button.

(b) Create a table PAY with the following fields :

EMPNO, NAME, DOJ, BPAY, DA, HRA, DED, GPAY, NPAY

(i) Add 5 records (except DA, HRA, DED, GPAY and NPAY)

(ii) Fill DA with 50% of BPAY for all employees

(iii) Fill HRA with 1500 for all employees

(iv) Fill DED with 750 for all employees

(v) Fill GPAY with the sum of BPAY, DA and HRA

(vi) Fill NPAY with GPAY-DED

(vii) Display the contents of the table.

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5. (a) Write and test a VB program to read principal, number of years, rate of interest through INPUT boxes. Compute and print the simple interest and compound interest through labels.

(b) Create a table ADDRESS with the following fields :

NAME, STREET, CITY, PIN

(i) Add 6 records and display the structure of the table

(ii) Add the field DIST

(iii) Fill all DIST with CHENNAI

(iv) Change the PIN 630030 where city is TRICHY and PIN is 628002.

DE–6615 **37**

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2013.

LAB V — JAVA PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examiner for each candidate and given to them by Lot system.

Each subdivision carries 50 marks.

1. (a) Write a program in Java to create multiple threads.

   (b) Write a Java program that displays the number of characters, lines and words in a text file.

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2. (a) Write a Java program to multiply the given two matrices.

   (b) Write a program to display the contents of file along with a line number before each line.

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3. (a) Write a program to create packages in Java.

   (b) Write a Java program to create and manipulate Text areas, Canvas, scroll bars, frames and menus.

4. (a) Write a Java to arrange the given names in alphabetical order.

   (b) Write a Java program to implement producer/consumer problem using synchronization.

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5. (a) Write a Java program which illustrates the implementation of multiple inheritance using interfaces.

   (b) Write a Java program to read and write the data using Random access file.

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6. (a) Write a program to create an applet and draw the various polygons.

   (b) Write a program in Java to count the numbers of words and characters in a text.

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7. (a) Write a Java program which illustrate the implementation of single and hierarchical inheritance.

   (b) Write a Java program that read a file name from the user and display the contents of file.

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8. (a) Write a Java program to illustrate the overload and overriding methods.

   (b) Write a Java program to create and manipulate labels, lists, text fields, text areas and panels.